

GenPod: Constructive News Framing in AI-Generated Podcasts More Effectively Reduces Negative Emotions Than Non-Constructive Framing

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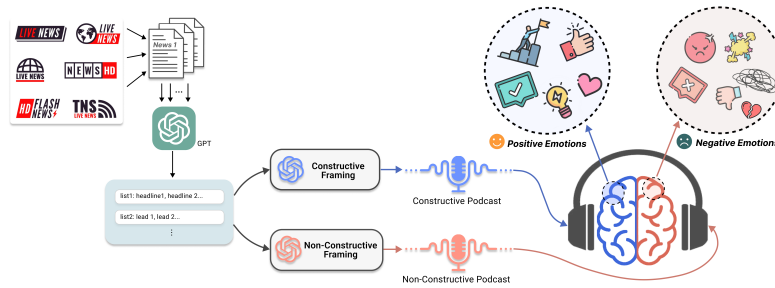


Figure 1: An illustration of the generative podcast production pipeline with constructive or non-constructive framing methods and their different influences on emotions. A more comprehensive process description is given in Section 4.

Abstract

AI-generated media products are increasingly prevalent, yet its emotional impact on audiences remains underexplored. This paper presents GenPod, the five-stage, modular generative pipeline that transforms identical news inputs into constructively or non-constructively framed podcast versions via prompt engineering and neural TTS. In a mixed-methods study ($N=65$), we found that constructive framing via GenPod significantly reduces negative affect compared to non-constructive framing. We further identify solution visibility, temporal framing, and tone balance as core mechanisms driving emotional shifts. Our findings highlight framing as a powerful design lever for creating more ethical and emotionally supportive AI-generated media.

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CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; Sound-based input / output; Interaction design theory, concepts and paradigms; • **Computing methodologies** → Natural language generation; • **Applied computing** → Media arts.

Keywords

Generative AI; Prompt Engineering; Podcast; News Framing; Emotions

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1 Background

As generative AI continues to advance, AI-generated news has emerged as a scalable tool for content production and personalization. Among these formats, AI-generated podcasts are gaining attention for their potential to automate content creation [42] and deliver tailored personalized information [17]. Despite growing

adoption, the emotional impact of such content remains underexplored, particularly how different framing styles might influence audience emotions [25]. This paper addresses that gap by examining the role of framing in shaping emotional engagement with AI-generated news podcasts.

1.1 AI in News and Podcasting

Generative AI has advanced automated news production [22, 28], enabling systems to draft articles, personalize content [15, 53], and streamline editorial workflows [43]. Yet ethical issues of AI-generated content have surfaced, such as bias [33, 50], misinformation [19], and misleading narratives [40, 52].

Podcasts, in particular, offer a multimodal platform for emotional storytelling, valued for their linguistic richness [29] and multitasking-friendly design [27, 35, 41]. AI-driven podcast systems can automate summarization [36] and recommendation [1, 54], but few go beyond surface-level customization to address how narrative structure influence listener affect. While efforts to improve content transparency and reliability continue [11, 23], the role of framing in news presentation remains underexplored [6, 10].

1.2 Impact of News Framing on Audience Cognition and Behavior

Media framing theory [12] suggests that how information is structured—via tone, emphasis, or narrative framing—can shape interpretation and emotional impact [24]. Negative framing [44], such as conflict-centered or sensationalized coverage [45], often elicits anxiety or outrage [37], increasing engagement but fostering long-term cynicism [2, 39]. In contrast, constructive journalism focuses on solutions, positive outcomes, and future perspectives [20], which may enhance resilience and proactive behavior [9, 31].

While framing effects have been studied in traditional media, little is known about how these dynamics play out in AI-generated audio content [4, 7]. Our study explores this gap by asking: **How do audience emotions differ when the same news resources are compiled into podcasts by an LLM using either constructive or non-constructive framing?**

To investigate this, we developed a generative podcast pipeline that takes identical news materials as input and, through prompt engineering and TTS synthesis, to produces both constructive and non-constructive framed podcast. We then evaluated their emotional effects in a controlled user study.

2 FORMATIVE STUDY

We conducted a formative study with two goals: (1) to gain a vivid understanding of listeners' current listening practices and (2) to obtain their specific expectations for generative podcasts.

We recruited 8 regular podcast listeners (F1-8, 4 female, 4 male; age, $M = 24.25$, $SD = 3.03$) in two remote workshops to inform our GenPod design. Participants first discussed their listening habits and then evaluated three 1.5–2 minute AI-generated podcast prototypes—monologic briefings, dyadic dialogues, and hybrid narratives—created using GPT-4o [34] and Azure TTS [32]. The following co-design activity had participants create their podcasts.

Through discussion, two clear preferences emerged: **(1) News podcasts were deemed an ideal format for AI Generation.**

Participants noted that news's structured, factual nature aligns with AI's strengths in objective processing: *"News exists to deliver real-time updates efficiently—AI can process stories way faster than any human ever could (F2)."* AI was also perceived as capable of delivering more objective perspectives, with *"AI won't downplay climate data because some CEO donated to a campaign (F6)."* They also noted that the typically neutral tone of news mitigates AI's emotional flatness: *"The tone is rather flat (F1),"* *"which works well for reporting emotion-neutral news (F4)."* **(2) Dyadic dialogues were considered more engaging for generative news podcast.** Conversational formats foster connection and clarity: *"AI can connect more emotionally with a dialogue, as it mimics real-life conversations (F2),"* The back-and-forth breaks down complex information: *"The other person repeats or asks questions, helping clarify the point (F6)."* This style was seen as reducing monotony and humanizing AI voices as *"it shifts attention away from the voice itself (F4)."*

These insights directly shaped GenPod's pipeline by focusing on structured news material and employing a two-host dialogue style.

3 GENPOD: A GENERATIVE AI PIPELINE FOR NEWS PODCASTS

To investigate how framing affects audience emotions in AI-generated audio, we developed GenPod, a multi-stage pipeline that produces constructive and non-constructive news podcasts from the same input sources. The system consists of five modular agents that sequentially process news material into finished podcasts using large language models and text-to-speech (TTS) synthesis. Fig. 2 illustrates the full workflow.

3.1 Dissecting News From News Sources

GenPod begins by dissecting input articles from news sources using **Agent 1**. Guided by a system prompt based on standard journalistic conventions, the model identifies five core components: headline, lead, body, background, and conclusion [38, 46]. This dissection preserves original content without modification and provides the foundation for controlled reframing. The detailed prompt structure and output of Agent 1 are shown in Fig. 3.

3.2 Recompiling News with Constructive and Non-Constructive Frames

To explore the emotional impact of framing, **Agent 2** and **Agent 3** recompile the same segmented content into constructive and non-constructive versions, respectively. Constructive news emphasizes solutions, agency, and future outlooks [30], while non-constructive news focuses on problems, conflict, and pessimism [2, 47].

The agents recompile news using structured lists, maintaining original details to preserve quality. Few-shot examples [5, 51] ensures that both reframings retain informational accuracy while shifting tone and perspective. Outputs are validated by **Agent 4**, which checks adherence to structural and stylistic criteria.

3.3 Podcast Scripting and Audio Production

Agent 5 transforms each reframed news version into a dyadic dialogue script, assigning framing-consistent personas to two AI

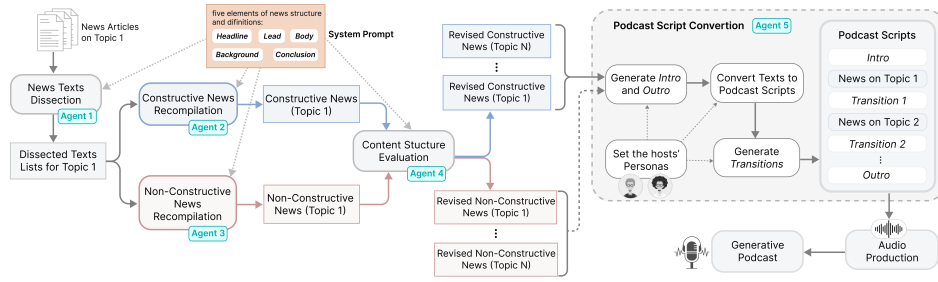


Figure 2: GenPod pipeline for generating constructive and non-constructive podcasts. Five sequential agents process news input: Agent 1 dissects content; Agents 2 and 3 apply different framings; Agent 4 validates structure; Agent 5 creates scripts for text-to-speech podcast production.

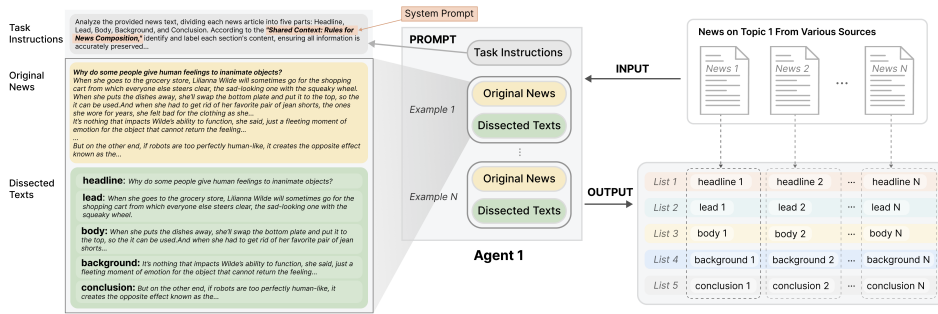


Figure 3: Agent 1 workflow and prompt design. News articles are segmented into five components—headline, lead, body, background, conclusion—based on predefined structure rules, forming the input for reframing.

hosts [16]. The final scripts are synthesized into audio using GPT-SoVITS beta0706 [21], neural TTS model optimized for realistic speech and voice cloning.

4 METHODOLOGY

The study investigates the emotional impact of podcasts generated from GenPod with constructive and non-constructive perspectives. A mixed-methods approach was employed, combining a between-subjects experiment with follow-up qualitative interviews.

Two podcast versions—constructive and non-constructive—were generated using the GenPod pipeline, each approximately 3 minutes and 45 seconds long. Both versions covered the same two news topics (food delivery labor and Olympic sports fandom), and were voiced, scored, and structured identically to ensure comparability. To ensure topic consistency and source credibility, we manually collected recent articles from established Chinese news outlets. Podcasts were published on a major Chinese platform ("Xiaoyuzhou"), accompanied by full transcripts.

4.1 Participants and Procedure

We recruited 66 participants (P1-66, 34 males, 31 females, 1 non-binary; aged 18-51, $M = 23.23$, $SD = 5.42$) through social media.

Each participant completed a pre-experiment survey including the 10-item Positive and Negative Affect Schedule (PANAS) [49]. They were then randomly assigned to listen to either the Constructive (CP) or Non-Constructive (NP) podcast version. Afterward,

they completed a comprehension check and a post-experiment PANAS to measure emotional changes.

Semi-structured interviews were conducted with a randomly selected subset of seven participants. Each interviewee listened to the alternative podcast version they had not encountered during the experiment, enabling comparative reflection. Interviews focused on emotional reactions and perceptions of framing, and were audio recorded, transcribed, and anonymized.

4.2 Data analysis

Quantitative data were analyzed using ANOVA to assess emotional changes across conditions. For interview data, we used an inductive thematic analysis approach [3]. Two researchers independently coded transcripts and iteratively refined the codebook to identify recurring themes related to emotional framing, such as solution visibility and tone balance.

5 FINDINGS

Of the 66 questionnaires collected, one was excluded at the comprehension check, leaving 65 valid participants. We present quantitative and qualitative findings comparing listener responses to the Constructive Podcast (CP) and Non-Constructive Podcast (NP).

Quantitative Results. ANOVA results (Fig. 4) revealed a significant reduction in negative emotions among participants who listened to CPs ($M = -2.36$, $SD = 4.46$) compared to those who listened to NPs ($M = 0.53$, $SD = 3.85$), $F(1, 63) = 7.815$, $p < 0.01$.

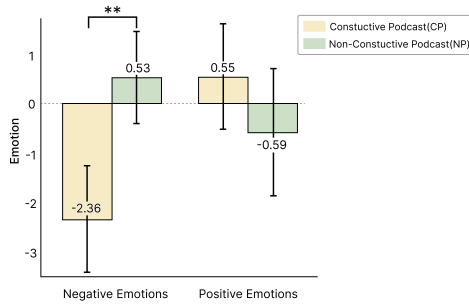


Figure 4: Results of the post-survey showed that constructive podcasts significantly reduced negative emotions on topic 1: food delivery workers. ** indicate significance of $p < 0.01$.

No significant difference was observed in positive emotions (CP : $M = 0.55$, $SD = 4.44$; NP : $M = -0.59$, $SD = 5.31$) with $p = 0.351$.

Qualitative Results. Thematic analysis of interview data revealed three key framing factors that shaped emotional responses: visibility of solutions, temporal framing, and tone balance.

Non-Constructive Podcasts (NPs) were frequently described as emotionally draining due to their narrow focus on severe issues. P3 noted, "about the delivery worker struggling to make ends meet, NP presented the situation as a dead-end... there's no way out." This effect was amplified by the lack of forward-looking context. P5 remarked, "NP made sports fandom sound toxic, with no hope for change."

Constructive Podcasts (CPs) provided a more balanced perspective by including progress or policy responses, which helped mitigate negative affect. P1 said, "CP didn't sugarcoat the struggles of people, but hearing about new laws gave it a sense of perspective." P6 described CPs as "a harsh but hopeful reality check," and P4 likened it to "seeing the fire but also being handed a fire extinguisher."

The tone of CPs was consistently described as calm, grounded, and motivational. P3 highlighted, "the success stories of other delivery workers made me feel like there was a path forward." In contrast, even when NPs mentioned solutions, participants found their pessimistic framing undermined any hope, as P6 noted, "NP emphasizes the negative consequences, even when discussing solutions."

6 DISCUSSION

Our findings show that constructive framing in AI-generated podcasts significantly reduces negative affect, while effects on positive affect were limited. This aligns with the broaden-and-build theory where increases in positive emotions tend to require longer or repeated exposure to take effect [14]. Still, qualitative responses suggest that constructive framing offers a perceptual shift toward hope, calm, and engagement.

6.1 Interpreting the Emotional Impact of Framing in AI-Generated News

This study provides empirical evidence that narrative framing in AI-mediated media influences emotional outcomes. Participants attributed emotional responses to three framing strategies in CP: solution visibility, temporal framing (i.e., showing progress over

time), and tone balance. These themes aligned across listeners mirrored our quantitative results. For example, CP's mention of policy responses helped convey a "path forward," while NP's lack of future framing evoked helplessness. Our results extend HCI literature on emotional design and interface framing to the context of generative media. While prior studies examined trust and stress in digital systems [8, 18], our findings show that affective responses also apply to voice-based AI content. AI-generated news, like traditional news, is not emotionally neutral—framing choices in prompts and voices shape user experience.

By operationalizing framing strategies in GenPod, we provide a replicable framework that future HCI researchers can adapt. This turns framing from a passive media bias into an explicit, testable design parameter with measurable emotional consequences.

6.2 Constructive Framing as a Path Toward Ethical AI Media

The rise of generative AI in content production has accelerated concerns about misinformation [19, 48], bias [50], and emotional manipulation [40] in automated media. While GenPod was not designed to combat misinformation directly, it introduces a lightweight but impactful design strategy to support user well-being without compromising factual integrity. Our findings suggest that AI-generated media can adopt editorial values such as transparency, civility, and solution orientation through framing decisions made at the design level. In this way, GenPod functions as a design probe for responsible media generation.

Prior work has shown that sensationalism in algorithmic content leads to disengagement and anxiety [26]. By contrast, participants in our study described CPs as emotionally manageable and motivational—even when discussing difficult topics. This supports the idea that framing is not simply a stylistic choice, but a determinant of emotional wellbeing and attention sustainability [13].

6.3 Limitations and Future Work

This study is limited by its short-form design and a relatively homogeneous participant pool. While results were consistent across listeners, emotional effects may vary by demographic background, topic sensitivity, or voice delivery style. Future work should explore longitudinal effects, personalized framing strategies, and applications across other media formats such as video. Further research is also needed to examine how constructive framing interacts with trust, comprehension, and behavioral outcomes, particularly in contexts like public health, crisis reporting, or civic engagement.

7 CONCLUSION

This work presents GenPod, an AI-generated pipeline that produces news podcasts with distinct framings from identical inputs. Our study shows that constructive framing effectively reduces negative emotions without altering factual content. The results demonstrate that framing in AI-generated media is a designable feature, and that incorporating constructive framing can support audience well-being. By operationalizing framing within the generative process, we offer a replicable method for designing more ethical and human-centered AI media systems.

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